

Cecilia Ferrando

☎ (+1) 412-636-1876 | ✉ cferrando@umass.edu | 🐙 Github | 🔗 LinkedIn | 📄 Google Scholar

Research interests: Statistical Machine Learning, Differential Privacy, Privacy-Preserving Machine Learning

Education

University of Massachusetts Amherst

Amherst, MA

PHD IN COMPUTER SCIENCE, MANNING COLLEGE OF INFORMATION AND COMPUTER SCIENCES

Aug. 2019 - Spring 2026 (exp.)

- Manning CICS Thesis Writing Fellowship recipient (2024)
- Dean's Outstanding Leadership Award recipient (2021)
- CS Fellow (merit scholarship for Computer Science students, 2019-2020)
- Cadence Women in Technology Scholar (2019-2020)
- GPA: 3.95/4

Carnegie Mellon University

Pittsburgh, PA

MS IN COMPUTATIONAL DESIGN, FOCUS ON MACHINE LEARNING

Aug. 2016 - May 2018

- Fulbright Scholar

Polytechnic University of Turin and Collegio Carlo Alberto

Turin, Italy

HONORS BS+MS DOUBLE DEGREE, MAJORS: ECONOMICS AND STATISTICS, ARCHITECTURE

Sep. 2010 - Jul. 2016

- Collegio Carlo Alberto "Allievi" Scholar
- Alta Scuola Politecnica Scholar
- Won EU-funded scholarship to study abroad in Paris, France (2014-2015)

Research Experience

LinkedIn

New York, NY

AI/ML PHD INTERN. MANAGER: LINDA FAYAD

May 2025 - Aug. 2025

- Designed and implemented LLM-based inference models for company-specific AI-driven learning tasks
- Designed evaluation frameworks for LLM inference tasks, including embedding-based metrics and LLM-as-a-judge frameworks
- Applied LLM inference models to real-world user data, demonstrating measurable inference improvements

Meta

New York, NY (remote)

RESEARCH ENGINEER INTERN, STATS & PRIVACY R&D TEAM. HOST: JAMES HONAKER, PHD

May 2022 - Aug. 2022

- Researched and applied novel statistically valid differentially-private inference methods for two company-specific problems
- Provided the engineering team with a ready-to-use implementation of my algorithms that the company can now adopt on multiple private inference pipelines

Google Research

New York, NY (remote)

RESEARCH INTERN. HOST: ALEX KULESZA, PHD

May 2021 - Aug. 2021

- Conducted research on novel differentially-private inference methods and supporting theory
- Independently designed and ran extensive experiments to validate new methods
- Formalized results into a conference workshop paper (accepted at PriML at NeurIPS 2021)

University of Massachusetts Amherst

Amherst, MA

RESEARCH ASSISTANT TO PROF. DANIEL SHELDON, PROBABILISTIC MACHINE LEARNING

Aug. 2019 -

- Developing new methods and algorithms for noise-aware differentially private inference and uncertainty estimation
- Published research papers on differentially private machine learning
- Implemented and published code to execute experiments validating my methods, including output visualization

Carnegie Mellon University

Pittsburgh, PA

RESEARCH ASSISTANT TO PROF. DANIEL CARDOSO LLACH, SPATIAL MACHINE LEARNING

May 2017 - Apr. 2018

- Published research on applications of machine learning for spatial analysis of architectural plans
- Presented Master's thesis work at Spatial Cognition 2018, winning Best Poster Presentation award
- Developed models to classify architectural plans based on their graph embedding
- Contributed to curating an exhibition on the origins of computer-aided design (CAD)

Other Experience

Cadence Design Systems

Pittsburgh, PA

MACHINE LEARNING SOFTWARE ENGINEER

Jun. 2018 - May 2019

- Independently led applied research on generative adversarial networks using capsule networks, implementing algorithms from scratch
- Improved the test performance of a classification algorithm by 11% by integrating capsule networks in the computer vision pipeline
- Invited to present my methods and results at Cadence Machine Learning Summit 2019, attendees included company's top scientific and executive leadership

Procore Technologies

Carpinteria, CA

QUANTITATIVE RESEARCH INTERN

May 2017 - Jul. 2017

- Developed data analytic app for uncertainty estimation of statistics derived from user data. The company has adopted my app for routine use in their survey data analysis pipeline
- Collaborated with UX, Engineering and Finance teams

Research Papers

C. Ferrando, D. Sheldon, *Private Regression via Data-Dependent Sufficient Statistic Perturbation*, TMLR 2025

C. Ferrando, S. Wang, D. Sheldon, *Parametric Bootstrap for Differentially Private Confidence Intervals*, AISTATS 2022

C. Ferrando, J. Gillenwater, A. Kulesza, *Combining Public and Private Data*, PriML Workshop at NeurIPS 2021

C. Ferrando, N. Dalmaso, J. Mai, D. Cardoso Llach, *Architectural Distant Reading - Using Machine Learning to Identify Typological Traits Across Multiple Buildings*, CUMINCAD 2019

Workshop Posters

C. Ferrando, D. Sheldon, *Private Regression via Data-Dependent Sufficient Statistic Perturbation*, TPDP 2024

C. Ferrando, D. Sheldon, *Parametric bootstrap for correcting clamping and truncation bias in differential privacy*, Women in Machine Learning Workshop, NeurIPS 2020

C. Ferrando, *A Machine Learning Framework for Spatial Analysis*, Best Poster Presentation award, Spatial Cognition 2018

Selected Projects

Spring 2021	2020 National Institute of Standards and Technology Differential Privacy Temporal Map Challenge, team won \$43,000 prize , with Ryan McKenna, Joie Wu, Arisa Tajima, Brett Mullins, and Siddhant Pradhan
Fall 2019	Adaptive robust regression for heteroskedastic data , UMass Amherst CS689 Machine Learning, final project with Kenta Takatsu
Spring 2017	HP-Intel NASA Design Challenge "Life in Space" , 1st prize winning team project
Fall 2016	The Harmonograph , Carnegie Mellon 15-112 Fundamentals of CS and Programming, final project, 2nd prize over 400+ projects

Honors & Awards

SCHOLARSHIPS AND FELLOWSHIPS

2024	Manning CICS Thesis Writing Fellowship , CICS, University of Massachusetts Amherst
2022	GHC Scholarship , Anita Borg, Grace Hopper Celebration
2020	Cadence Women in Technology Scholarship , Cadence Design Systems
2019	CS Fellowship , CICS, University of Massachusetts Amherst
2016-2018	Fulbright Scholarship (\$40,000) , US Department of State
2010-2016	Collegio Carlo Alberto "Allievi" Scholarship (\$17,000) , Collegio Carlo Alberto, Turin, Italy
2013-2015	Alta Scuola Politecnica , top 1% students, Polytechnic University of Turin, Italy
2014-2015	Erasmus+ Scholarship , European Union

AWARDS

2021	Dean's Outstanding Leadership Award , for piloting the PhD Applicant Support Program, CICS, University of Massachusetts Amherst
2021	2020 NIST Differential Privacy Temporal Map Challenge (\$43,000) , with the Minutemen team
2020	NeurIPS 2020 registration award , Women in Machine Learning
2020	GHC 2020 registration award , CICS, University of Massachusetts Amherst
2020	ICML and ICLR 2020 registration award , Women in Machine Learning
2019	NeurIPS 2019 travel and registration award , NeurIPS Conference
2017	First Prize , HP-Intel "Life in Space" Design Challenge, CMU team
2017	Second Prize , Carnegie Mellon 15-112 best CS projects over more than 400
2009	Albo delle Eccellenze , top high-school students in Italy, Italian Ministry of Education

CONFERENCE PRESENTATIONS

2018	Spatial Cognition 2018 Best Poster Award , Spatial Cognition 2018, Tuebingen, Germany
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Teaching & Mentorship

University of Massachusetts Amherst

Amherst, MA

TEACHING ASSISTANT, CS119 INTRODUCTION TO PROGRAMMING

Fall 2023, Spring 2024, Spring/Fall 2025

- Contributed to course organization, office hours, assignments and grading
- Organized hands-on programming labs for the students
- Rated 'Outstanding' in TA evaluations

University of Massachusetts Amherst

Amherst, MA

GRADUATE THESIS MENTOR TO UNDERGRADUATE STUDENT ADI GEVA

Feb. 2022 - May 2022

- Mentored Adi as she worked on her thesis project on bootstrap methods for differentially private confidence intervals. Adi won the UMass Manning CICS 2022 Outstanding Undergraduate Achievement Award

University of Massachusetts Amherst

Amherst, MA

GRADUATE MENTOR, CICS EMBER PROGRAM

Feb. 2021 - May 2021

- Mentoring undergraduate students from underrepresented groups conducting applied research on uncertainty quantification in differential privacy

University of Massachusetts Amherst

Amherst, MA

GRADUATE MENTOR, CICS UNDERGRADUATE RESEARCH PROGRAM

Jun. 2020 - Jan. 2021

- Mentoring five undergraduate students conducting applied research on uncertainty quantification in differential privacy
- Provided weekly 1-1 guidance and feedback tailored on each student

Service & Leadership

2022-	NeurIPS, ICML, ICLR, AISTATS, JMLR , Reviewer
2020-2022	PhD Applicant Support Program (PASP) , UMass Amherst CICS, Co-Chair. A mentorship program supporting prospective PhD students. Received Dean's Outstanding Leadership Award.
2020-2021	Voices of Data Science , Co-Chair of the inaugural Voices of Data Science at UMass Amherst conference.
2020	UMass Graduate CS Women group , Social Co-Chair. Organized networking events for CS women graduate students and faculty.
2020	New Student Committee , UMass Amherst CICS. Contributed to PhD candidate visit day.

Skills

Programming	Python (NumPy, SciPy, Pandas, PyTorch, Opacus, TensorFlow, TensorFlow Privacy, JAX, Jupyter, Matplotlib, Seaborn), Cloud (AWS), C++, Matlab, Git, LaTeX
Design	Adobe Photoshop, InDesign, Illustrator, Lightroom, PremierePro. Autodesk AutoCAD, 3DSMax. Rhino, Grasshopper
Languages	Italian (native), English (advanced), French (advanced)